

Mid Semester Examination 2024

Name of the Program: B.Tech
Name of the Course: Electrical Drives

Semester: 7th
Course Code: **TEE701**

Time: 3 Hour

Maximum Marks: 100

Note:

- (i) Answer **all the questions** by choosing **any two of the sub questions**.
- (ii) Each question carries 20 marks

Q1	(20 marks)	
(a)	Draw the various plots for steady state stability limits of electrical drives	CO1,CO3
(b)	A variable speed d.c. drive has rated power of 10 kW, rated speed of 1500 rpm drives a load that comprises a constant load of $T_L = 30 \text{ Nm}$. The inertia of the drive system is 0.10 kg.m^2 . Calculate the time taken to accelerate the load from zero to 800 rpm , assuming the drive develops rated torque during the acceleration phase.	
(c)	What is regenerative braking in a DC motor, and when is it used? Discuss with suitable diagram	
Q2	20 marks	
(a)	Discuss the multi-quadrant operation of electrical drives	CO2,CO3
(b)	What are the main performance characteristics of a DC motor?	
(c)	How is the starting of a DC motor typically controlled, and why is it necessary?	
Q3	(20 marks)	
(a)	Discuss the role of power modulator for electrical drives	CO3,CO5
(b)	What are the common types of braking methods used for DC motors? Discuss	
(c)	Draw the block diagram of closed loop drive system and explain its various component	
Q4	(20 marks)	

(a)	A drive used in a hoist to raise and lower weights up to 400 kg at velocities up to ± 2 m/s. The weight hangs from a cable that is wound on a drum of radius of 0.4 m . The drum is driven by the drive motor through a gearbox that has an efficiency of 85% . The maximum speed of the motor is ± 1300rpm. It is required to: Sketch the system and find the nearest integer gearbox ratio that will match the maximum speed of the motor to the maximum velocity of the hoist. Determine the torque and power provided by the motor when lifting the maximum weight at the maximum velocity. (c) Calculate the torque and power provided by the motor when lowering the maximum weight at the maximum velocity.	CO6.CO1
(b)	Draw block diagram of electrical drives and explain its various parts	
(c)	Explain the working of three phase fully controlled rectifier used for dc drives	
Q5	(20 marks)	
(a)	Discuss the load torque equation considering frictional, Windage, static friction and coulomb friction.	CO6,CO5
(b)	A constant speed drive has the following duty cycle: (i) Load rising from 0 to 400 kW : 5 min (ii) Uniform load of 500 kW : 5 min (iii) Regenerative power of 400 kW returned to the supply : 4 min (iv) Remains idle for : 2 min Estimate power rating of the motor. Assume losses to be proportional to (power)².	
c)	Discuss the slip power recovery scheme used for ac drives	